

Optimization-based machine learning for computational imaging

Conclusion

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Unified framework

Inference framework: feed-forward NN

$$\begin{aligned} (\forall \mathbf{x}^{[0]} \in \mathbb{R}^{N_0}) \quad \mathbf{x}^{[K]} &= \mathbf{f}_{\Theta}^K(\mathbf{x}^{[0]}) \\ &= \mathcal{T}_{\Theta_K} \circ \dots \circ \mathcal{T}_{\Theta_1}(\mathbf{x}^{[0]}), \end{aligned}$$

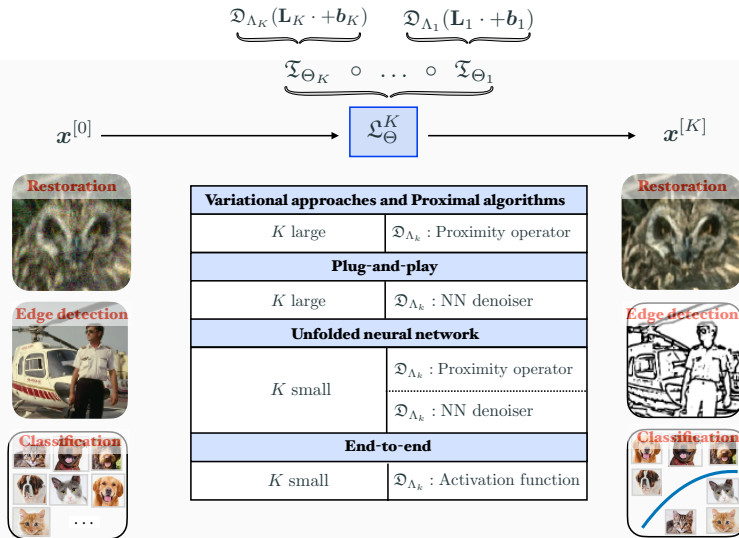
Layer/iteration

$$\mathcal{T}_{\Theta_k} : \mathbb{R}^{N_{k-1}} \rightarrow \mathbb{R}^{N_k} : \mathbf{x} \mapsto \mathcal{D}_{\Lambda_k}(\mathbf{L}_k \mathbf{x} + \mathbf{b}_k),$$

- $\mathbf{L}_k : \mathbb{R}^{N_{k-1}} \rightarrow \mathbb{R}^{N_k}$: linear operator,
- $\mathbf{b}_k \in \mathbb{R}^{N_k}$: shift parameter,
- $\mathcal{D}_{\Lambda_k} : \mathbb{R}^{N_k} \rightarrow \mathbb{R}^{N_k}$: nonlinear operator parametrized by Λ_k .

Parameters: $\Theta = \cup_{k=1}^K \Theta_k$ with $\Theta_k = \{\Lambda_k, \mathbf{L}_k, \mathbf{b}_k\}$.

Unified framework: Unfolded neural networks



Challenges for the next years

Theoretical challenges:

- Interpretation of output of (unfolded) neural networks
- Develop mathematical framework to better assess robustness of (unfolded) neural networks

Computational challenges:

- Boost expressivity of PnP and unfolded methods
- Further explore real applications

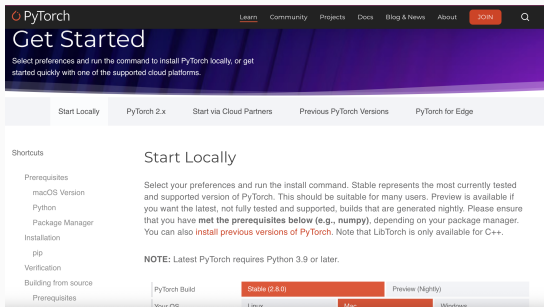
Societal challenges:

- Convince end-users that model-informed deep learning methods such as PnP and unfolded networks are reliable for decision-making processes
- Develop effective quantification measures for environmental impact of data-driven methods

Python toolbox

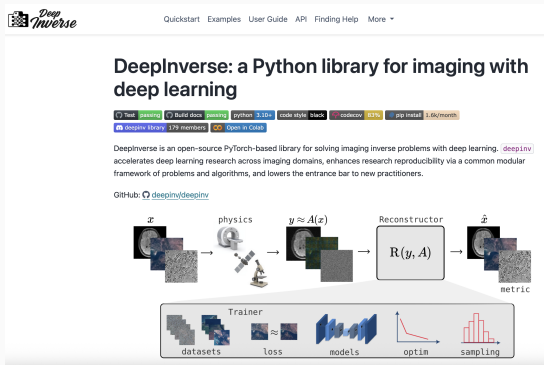
Python Toolbox: Based on DeepInverse and Pytorch

Goal: Find $\hat{\Theta} \in \Theta \frac{1}{|\mathbb{I}|} \sum_{j \in \mathbb{I}} \ell(\bar{x}_j, f_{\Theta}^K(z_j))$.



Python Toolbox: Based on DeepInverse and Pytorch

Goal: Design f_{Θ}^K



Python Toolbox: Based on DeepInverse and Pytorch

Goal: Design f_{Θ}^K

The screenshot shows the DeepInverse website interface. On the left is a 'Section Navigation' sidebar with categories like Basics, Models, Physics, Optimization, Plug-and-Play, Diffusion & MCMC, Unfolded, Self-Supervised Learning, Adversarial Learning, and External Libraries. The 'Basics' section is expanded, showing a '5 minute quickstart tutorial' as the first item. The main content area has a breadcrumb trail: 'Examples > Basics > 5 minute quickstart tutorial'. Below this is a 'Note' box stating: 'New to DeepInverse? Get started with the basics with the 5 minute quickstart tutorial.' The main heading is '5 minute quickstart tutorial', followed by the text 'Follow this example to get started with DeepInverse in under 5 minutes.' Below this is a 'Contents' list with links for 1. Install, 2. Physics, 3. Models, 4. Datasets, and 5. What's next. The '1. Install' section is currently selected and expanded, showing the text 'First, install and import the latest stable release of `deepinv`:'.

DeepInverse

Quickstart [Examples](#) User Guide API Finding Help More ▾

🔍 Search 🔍

🏠 > Examples > Basics > 5 minute quickstart tutorial

Section Navigation

- Basics
 - 5 minute quickstart tutorial
 - Use a pretrained model
 - Use iterative reconstruction algorithms
 - Bring your own dataset
 - Bring your own physics
- Models
- Physics
- Optimization
- Plug-and-Play
- Diffusion & MCMC
- Unfolded
- Self-Supervised Learning
- Adversarial Learning
- External Libraries

Note

New to DeepInverse? Get started with the basics with the [5 minute quickstart tutorial](#).

5 minute quickstart tutorial

Follow this example to get started with DeepInverse in under 5 minutes.

Contents

- [1. Install](#)
- [2. Physics](#)
- [3. Models](#)
- [4. Datasets](#)
- [5. What's next](#)

1. Install

First, install and import the latest stable release of `deepinv`:

On this page

1. Install
2. Physics
3. Models
4. Datasets

What's next?

This Page

- [Show Source](#)

[Download source code](#)

[Download Jupyter notebook](#)

[Download zipped](#)

Python Toolbox for model-based NN

<https://perso.ens-lyon.fr/nelly.pustelnik/PNN/>

Model-based neural networks

Proximal algorithms

Plug-and-Play

Unfolded

Contacts

This webpage provides basic codes to perform image reconstruction tasks with [DeepInverse library](#). To install it, follow the instructions provided [at this link](#).

We provide the codes necessary to reproduce part of the experiments detailed in our EUSIPCO tutorial "Proximal Neural Networks: Wedding Variational Methods and Artificial Intelligence" and additional codes to deepen the understanding. Utilizing the DeepInverse library enables us to concentrate on the structure of iterative schemes while leveraging established knowledge about data-term and prior design, as well as their associated gradient and proximity operators.

Proximal algorithms

- **Forward-backward**
 - Example in image restoration (Blur + Gaussian noise) with TV-L12 denoiser. [\[Code Python\]](#) [\[Notebook Google Colab\]](#)
- **FISTA**
 - Example in image restoration (Blur + Gaussian noise) with TV-L12 denoiser. [\[Code Python\]](#) [\[Notebook Google Colab\]](#)
- **Douglas-Rachford**
 - Example in image restoration (Blur + Gaussian noise) with TV-L12 denoiser. [\[Code Python\]](#)
- **Loris-Verhoeven**
 - Example in image restoration (Blur + Gaussian noise) with TV-L12 denoiser. [\[Code Python\]](#)
- **Condat-Vu**
 - Example in image restoration (Blur + Gaussian noise) with TV-L1 denoiser. [\[Code Python\]](#)
 - Example in image restoration (Blur + Gaussian noise) with TV-L12 denoiser. [\[Code Python\]](#) [\[Notebook Google Colab\]](#)
- **Chambolle-Pock**
 - Example in image restoration (Blur + Gaussian noise) with TV-L12 denoiser. [\[Code Python\]](#)

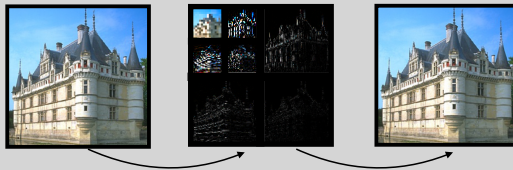
Plug-and-play

https://github.com/Nils-Laurent/multilevel_deepinv

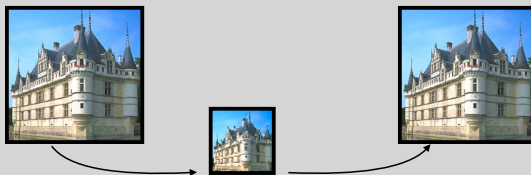
- ML-FB
- ML-FISTA
- ML-PnP

Focus on multiresolution / multilevel / multiscale

Multiresolution
to perform
image
restoration
(~2000–2015)



Multiresolution
to **accelerate**
algorithms
(~2016- now)



Multiresolution
to perform
texture
segmentation
(~2014- now)

